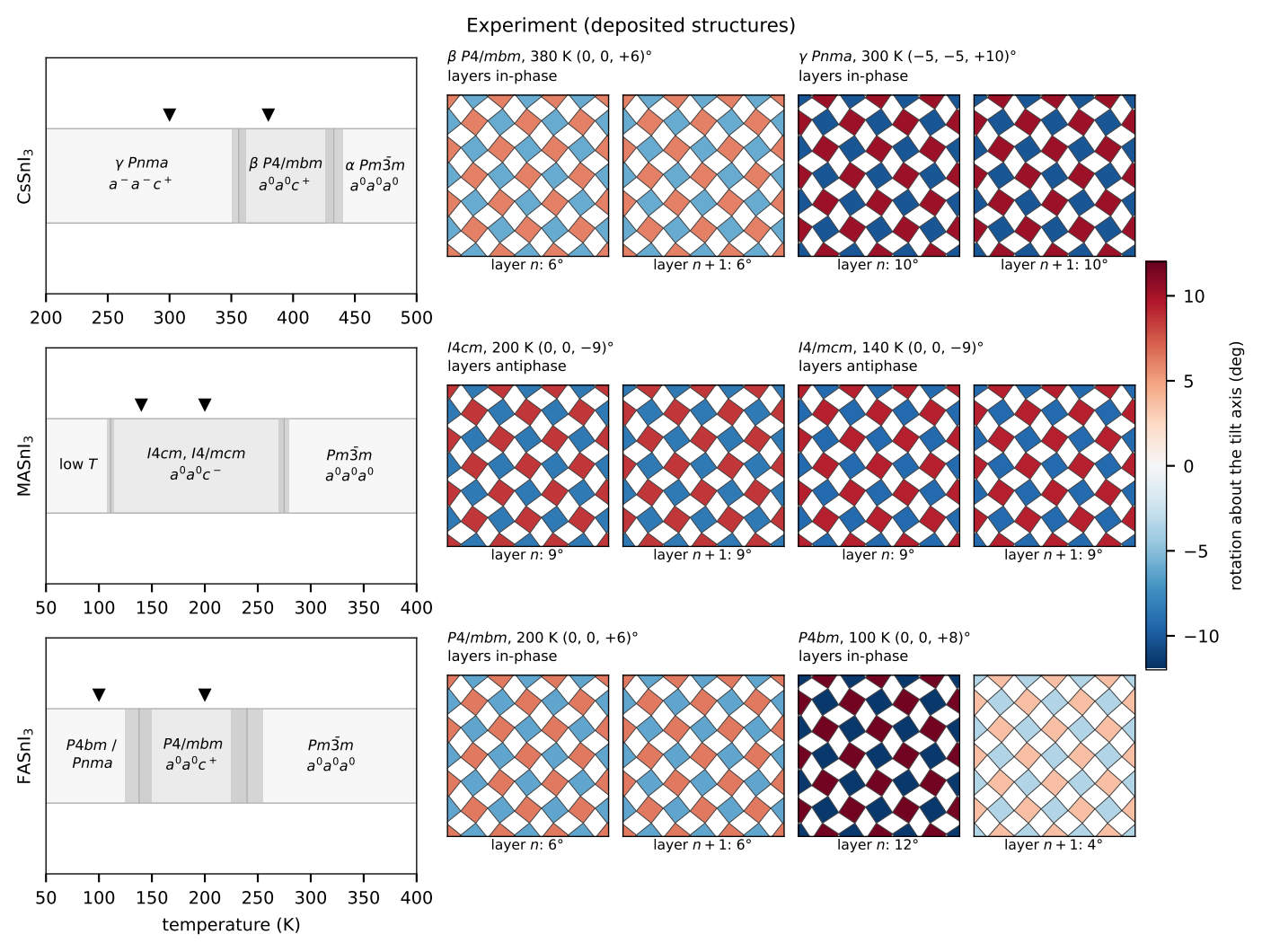
Experiment (deposited structures)
CsSnI3
γ Pnma
a⁻a⁻c⁺
β P4/mbm
a⁰a⁰c⁺
α Pm3̄m
a⁰a⁰a⁰
200 250 300 350 400 450 500
MASnI3
low T
I4cm, I4/mcm
a⁰a⁰c⁻
Pm3̄m
a⁰a⁰a⁰
50 100 150 200 250 300 350 400
FASnI3
P4bm / Pnma
P4/mbm
a⁰a⁰c⁺
Pm3̄m
a⁰a⁰a⁰
50 100 150 200 250 300 350 400
temperature (K)
β P4/mbm, 380 K (0, 0, +6)°
layers in-phase
layer n: 6°
layer n + 1: 6°
γ Pnma, 300 K (−5, −5, +10)°
layers in-phase
layer n: 10°
layer n + 1: 10°
I4cm, 200 K (0, 0, −9)°
layers antiphase
layer n: 9°
layer n + 1: 9°
I4/mcm, 140 K (0, 0, −9)°
layers antiphase
layer n: 9°
layer n + 1: 9°
P4/mbm, 200 K (0, 0, +6)°
layers in-phase
layer n: 6°
layer n + 1: 6°
P4bm, 100 K (0, 0, +8)°
layers in-phase
layer n: 12°
layer n + 1: 4°
10
5
0
−5
−10
rotation about the tilt axis (deg)